

MICROBIOLOGY

Lecture 1: Meningitis I

Meningitis: It's an infection and inflammation of the membranes surrounding the brain and spinal cord.

- Most causes are **infectious** and much less ones are **non-infectious** (such as medications, SLE, sarcoidosis, and carcinomatosis).
- If the course lasts longer than 4 weeks, it's defined as **chronic meningitis**
- knowing whether a virus or a bacterium causes meningitis is important because the severity of illness and the treatment differ.
- **Bacterial meningitis** can be severe and may result in brain damage, hearing loss, or learning disability. Thus, bacterial meningitis is a **medical emergency** that indicates prompt initiation of antibiotic treatment.
- Viral meningitis is generally less severe and resolve without specific treatment.

There are two major forms of meningitis. septic and aseptic:

- **Septic meningitis** is typically caused by bacteria and CSF test result is positive with routine bacterial culture. If not treated, it may lead to lifelong disability or death.
- **Aseptic meningitis** can be caused by viruses, fungi, mycobacterium tuberculosis or spirochaetes. CSF test result is negative with routine bacterial culture.

Infectious causes of meningitis:

➤ Bacterial causes:

1. Streptococcus pneumonia (Pneumococcus): common for all patients

beyond the neonatal period.

2. *Neisseria meningitidis* (Meningococcus): common in adolescents and spread by respiratory droplets.

3. *Haemophilus. Influenzae* (Hib): A very common cause in children in the past but now markedly decreased by the use of the Hib vaccine in children

4. Strep-B or *E. coli*: common in new-borns to infants of 6 months in age.

Listeria and *Klebsiella* are also common for this age group.

5. *Listeria*: More common in immunodeficient (particularly T cells or neutrophil deficiency) or immunocompromised.

6. *Mycobacterium tuberculosis* and spirochaetes.

7. Staph-aureus: More common with a history of neurosurgery.

➤ **Fungal cause** (*Cryptococcus*): More common in patients with markedly low T cells (as in AIDS).

➤ **Viral causes**: Enteroviruses (coxsackieviruses), herpes simplex type 2, mumps, west Nile virus and cytomegalovirus (CMV).

Morphological characteristic:

Pneumococcus: gram positive capsulated diplococci

Meningococcus: gram negative diplococci

Haemophilus. Influenzae: small pleomorphic gram-negative bacilli.

Streptococcus-B: gram positive capsulated cocci in chains

Listeria: short gram positive non spore forming rods

Mycobacterium tuberculosis: acid fast bacilli

Staph-aureus: gram positive cocci arranged in clusters

Cryptococcus (fungus): budding yeast cells surrounded by gelatinous capsule

Herpes simplex type 2, CMV: DNA enveloped virus

Enteroviruses (coxsackieviruses), mumps and west Nile virus: RNA viruses

Mode of infection:

1. Streptococcus pneumoniae (Pneumococcus): is transmitted by respiratory droplets
2. Neisseria Meningitidis (Meningococcus): is transmitted by droplets
3. Haemophilus. Influenzae: is transmitted by droplets
4. Mycobacterium tuberculosis: is transmitted by inhalation of droplet nuclei carried by air from the patient with open pulmonary tuberculosis
5. Cryptococcus: is transmitted by inhalation of fungi which present in soil contaminated with excreta of bird
6. Herpes simplex type 2 virus: is transmitted by sexual intercourse
7. Coxsackieviruses: are transmitted by oral-faecal route and by respiratory droplet

Pathogenesis

The organisms i.e. bacteria, virus or fungus that cause meningitis can gain access to the CNS and cause meningeal disease via any of the 3 pathways, invasion of blood stream, retrograde neuronal pathway and direct contiguous spread e.g., otitis media, trauma or sinusitis.,

The CSF is an ideal medium for the spread of bacteria because it provides enough nutrients for their multiplication and has few phagocytic cells, and low levels of antibodies and complement.

Clinical picture of meningitis:

The classic triad:

- Fever
- Headache
- Neck stiffness

Other symptoms: Altered mental status, nausea, vomiting, confusion, sleepiness

and coma

Prophylaxis and vaccination

Respiratory isolation and Rifampin or ceftriaxone for all close contacts of patients with meningococcus.

(2) MCV (meningococcal vaccine) is now recommended to all children at 11-12 years. Add a booster at 16 years or in asplenic or immunocompromised patients

(3) Haemophilus influenza type b vaccine: It provide protective immunity in children younger than 24 months

(4) BCG vaccine: It protect against childhood TB meningitis.